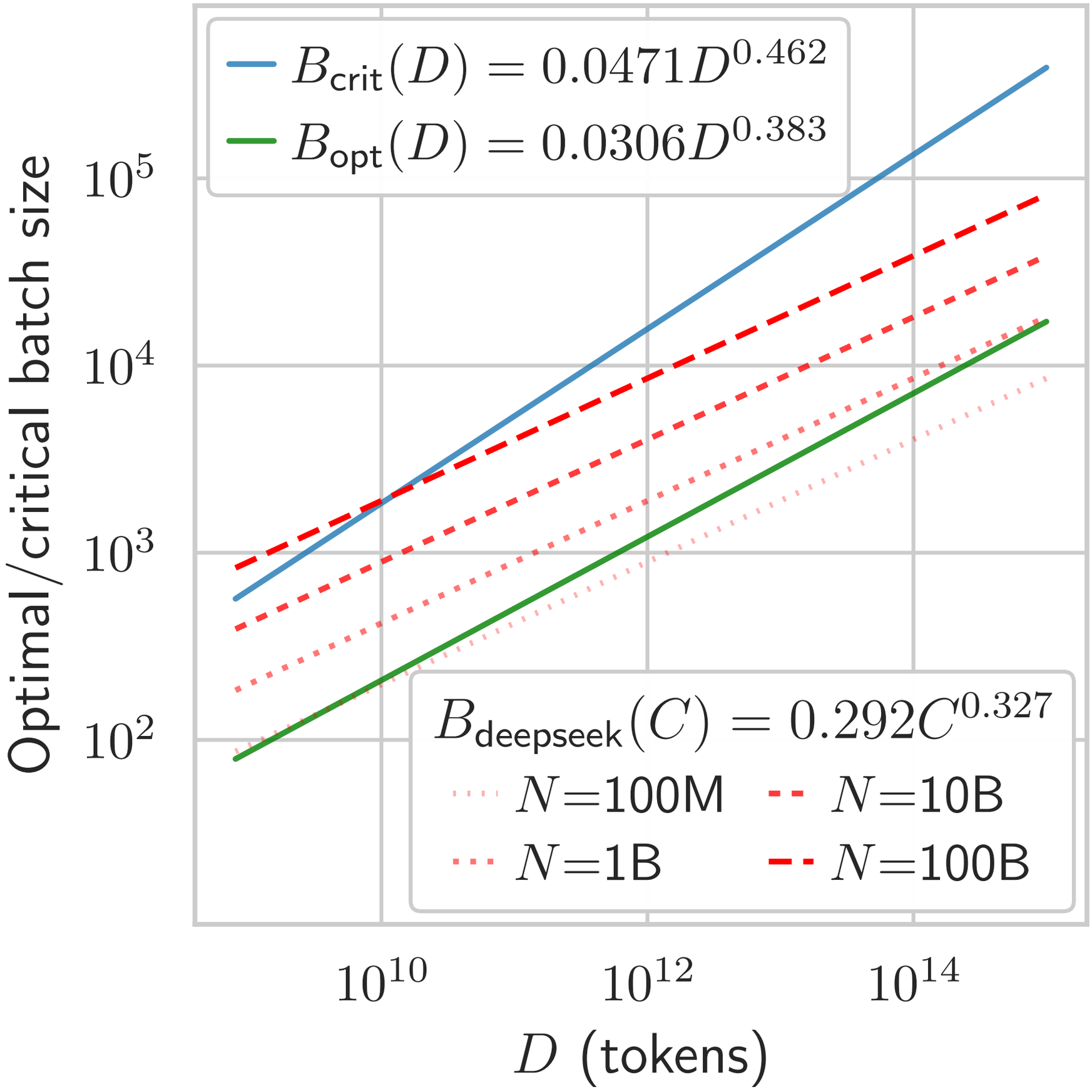

$B_{\text{crit}}(D) = 0.0471D^{0.462}$
$B_{\text{opt}}(D) = 0.0306D^{0.383}$
$B_{\text{deepseek}}(C) = 0.292C^{0.327}$
N=100M
N=10B
N=1B
N=100B
Optimal/critical batch size
10^5
10^4
10^3
10^2
10^{10}
10^{12}
10^{14}
D (tokens)